Prachatos Mitra

Email: pmitra32@gatech.edu **Phone**: +1 (404)-259-3077
Linkedin: `https://www.linkedin.com/in/prachatos/` **Website:** `https://prachatos.github.io`

EDUCATION

Georgia Institute of Technology January 2023 - Present
PhD, Computer Science GPA: 4.0/4.0
Advisor: Dr. Alexandros Daglis
Research Area: Simulators for large-scale systems, graph processing architectures

Indian Institute of Science August 2018 - July 2020
M.Tech., Computer Science GPA: 8.80/10.00
M.Tech thesis advisor: Dr. Arkaprava Basu
Research Area: Memory systems, TLB and cache replacement policies

Maulana Abul Kalam Azad University of Technology August 2014 - July 2018
B.Tech., Computer Science and Engineering GPA: 9.01/10.00

RESEARCH PROJECTS

- **Design and simulation for actor-model based sparse graph processing architecture**
  Georgia Institute of Technology 2023-2025
  - Captured algorithmic properties via fitted distributions to drive message schedules.
  - Built a stochastic SST simulator that scales to full-system runs (millions of PEs).
  - Used simulator for several case studies to select parameters for target architecture
  - Paper under review
- **Dead Page and Dead Block Predictors: Cleaning TLBs and Caches Together**
  Indian Institute of Science (Master's thesis) 2019-2020
  - Demonstrated the presence of "dead" or unused entries in the L2 TLB
  - Designed a dead-page predictor to improve TLB efficiency by bypassing such entries
  - Designed a dead-block predictor for last level caches using TLB info
  - Showed performance gains of over 8% with low storage overheads

PUBLICATIONS

[1] **HPCA-27:** Chandrashis Mazumdar*, **Prachatos Mitra***, Arkaprava Basu, "Dead Page and Dead Block Predictors: Cleaning TLBs and Caches Together", Proceedings of 27th IEEE International Symposium on High-Performance Computer Architecture (HPCA-27) (virtual), February, 2021. Available at: `https://bit.ly/3u9MwGU`

[2] **SC-W '23: Prachatos Mitra**, Alexandros Daglis, "Filtering Wasteful Vertex Visits in Breadth-First Search", Proceedings of the SC '23 Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis, November 2023. Available at: `https://doi.org/10.1145/3624062.3625133`

TALKS AND PRESENTATIONS

[1] **KVM Forum 2021: Prachatos Mitra** and Soham Ghosh "Support for Fast and Reliable VMM Live Upgrades in Libvirt", KVM Forum 2021 (virtual). Available at: `https://bit.ly/38cQHJT`

*__Equal contribution__

WORK EXPERIENCE

AMD Bangalore, India
Senior Silicon Design Engineer September 2022 - December 2022

Worked in Server Performance Group.

Conducted performance analysis and optimizations for virtualization and networking workloads on AMD server CPUs.

Nutanix Bangalore, India
Member of Technical Staff August 2020 - September 2022

Developed KVM- and qemu-based components for Nutanix AHV hypervisor. Contributed to control plane and live upgrade mechanisms via Libvirt.

TECHNICAL SKILLS

Languages	C/C++, Python
Simulation tools	SST, gem5, Sniper
Others	Libvirt, Qemu, KVM, Bash, LaTeX